

Sepehr Zhand *Ph.D.*

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Geospatial Data Scientist specialising in scalable spatial analytics and data engineering across multi-source geospatial environments. I design Python and PostGIS workflows that integrate complex spatial datasets, including network, demographic, and environmental data, to produce reproducible analytical systems and decision-ready insights. My work spans research, consultancy, and policy-facing projects, translating advanced spatial modelling into operational tools used for urban and infrastructure decision-making. I am particularly interested in applying spatial data systems to climate, environmental, and large-scale observational datasets.

Skills

Programming & Data Science	- Python (NumPy, Pandas, SciPy, scikit-learn) - Feature engineering and predictive modelling - Statistical learning and clustering
Geospatial & Spatial Analytics	- GeoPandas, Shapely, OSMnx, Fiona, Rasterio - Network analysis and spatial statistics - Vector and raster spatial processing - Multi-source spatial data integration
Data Engineering & Infrastructure	- PostgreSQL/PostGIS, SQL - Automated data pipelines (Python-based ETL) - Git / version-controlled workflows - Google Cloud Platform - Data validation and reproducibility practices
Visualisation & Communication	- Matplotlib - Spatial diagnostic reporting for decision support

Experiences

Associate – Spatial analyst | Studio Bednarski ltd. London, UK **Jan 2019 - Present**
Urban analytics, Computation and Insights

- Designed Python-based spatial analytics workflows integrating multi-source datasets (OSM, land-use, topography, demographic indicators) to support urban redevelopment and infrastructure planning.
- Built reproducible geospatial pipelines using GeoPandas and PostGIS to generate network and accessibility metrics across complex urban systems.
- Translated spatial modelling outputs into decision-support diagnostics used by planning and design teams.
- Led technical spatial analysis components across multidisciplinary projects, ensuring data consistency and methodological rigor

Research Fellow | University College London, UK
TWIN2EXPAND – Spatial Data Scientist

March 2023 - Dec 2025

- Designed and deployed automated spatial data pipelines covering 617 European urban areas using Python (OSMnx, Cityseer) and PostGIS on Google Cloud.
- Engineered multi-scale spatial features integrating network, demographic, and environmental indicators for downstream modelling.
- Implemented reproducible workflows for large-scale spatial dataset generation and validation.
- Produced structured datasets now used in ongoing urban analytics research and policy applications.

Research Associate | London School of Economics, UK
Roads as tools for (dis)connecting neighbourhoods – Spatial Data Scientist

May 2021 - June 2023

- Directed quantitative socio-spatial modelling across large-scale urban systems (~200k+ network nodes).
- Built predictive and clustering models integrating land-use, deprivation, and network structure datasets.
- Developed integrated analytical datasets supporting policy research outputs and publications.

Project consultant | Space Syntax Ltd., London UK
Geospatial analyst

May - Sep 2019

- Developed a bespoke land-value analytics tool for Melbourne by modelling a 500k+ node urban network and integrating multi-source spatial data. Processed OSM street geometry with Python (OSMnx, GeoPandas, Fiona, Shapely) and used DepthmapX and QGIS to derive accessibility indicators linked to land-use, traffic-flow, and POI datasets. Produced an integrated value surface that the municipality adopted as a decision-support tool for long-term planning.

Selected Project

Non-Linear Urban Dynamics | PhD project | UCL Space Syntax Lab

Sep 2018 - Feb 2025

- Developed a scalable spatial analytics framework for modelling non-linear urban dynamics across large-scale street-network systems (1M+ nodes). Built automated feature engineering workflows integrating multi-source spatial datasets and applied machine learning methods to estimate missing socio-spatial attributes.

Education

Ph.D. | Urban and Spatial analytics, Space Syntax Laboratory, UCL, UK

Sep 2018 - Feb 2025

MArch | Housing and Urbanism, AA School of Architecture, UK

Sep 2011 - March 2013

BArch | School of Architecture, University of Tehran, Iran

Sep 2006 - March 2011